SWA-MPPI: Socially-Weighted Alignment with Model Predictive Path Integral Control for Cross-Cultural Moral Alignment of Large Language Models

Anonymous Authors
anonymous@institution.edu

Abstract

Large language models inherit the moral intuitions of their predominantly Western training data, yet serve culturally diverse populations worldwide. No existing method can adapt an LLM's moral reasoning to a target culture *at inference time* without labelled preference data or weight modification. We introduce **SWA-MPPI**, the first training-free framework for cross-cultural moral alignment that operates entirely in logit space. SWA-MPPI constructs an ensemble of culturally grounded persona agents from World Values Survey data, detects inter-persona disagreement as a signal of cultural contestation, and resolves it via a Prospect-Theory-weighted Model Predictive Path Integral (MPPI) update—a principled stochastic optimisation that balances private cultural utility with collective welfare. On the Multilingual Trolley Problem benchmark across 15 countries, 12 native languages, and 6 moral dimensions, SWA-MPPI reduces Jensen-Shannon Distance to human moral preferences by **37.6%** (Qwen2.5-72B) and **34.7%** (Llama-3.1-70B), outperforming all prompting and activation-steering baselines by $\geq$10% while operating on frozen weights with a single additional batched forward pass. Notably, the largest gains occur for Anglosphere countries (+72–75%), revealing that RLHF-induced biases actively worsen alignment even for the best-represented populations. The framework is self-limiting: when the model is already well-calibrated, inter-persona consensus is high and no correction is applied. Code available at [repository].

1 Introduction

As large language models are deployed across increasingly diverse cultural contexts, a fundamental tension has emerged between the uniformity of a single trained model and the plurality of human moral values worldwide. Models trained predominantly on English-language corpora tend to inherit the ethical intuitions of Western, educated, industrialised, rich, and democratic (WEIRD) populations [Henrich et al., 2010], yet they are increasingly enlisted to assist with consequential decisions in societies whose moral norms differ substantially [Awad et al., 2018, Jin et al., 2025].

The Moral Machine experiment [Awad et al., 2018] collated over 40 million human judgments from 233 countries on trolley-problem scenarios, demonstrating profound cross-cultural variation in moral priorities. The multilingual successor benchmark, MultiTP [Jin et al., 2025], operationalises this evidence into country-specific Average Marginal Component Effects (AMCE) across six moral dimensions and 107 languages. Against this benchmark, Jin et al. [2025] found that none of 19 evaluated contemporary LLMs achieved broadly satisfactory cross-cultural alignment.

Existing approaches each sacrifice critical properties. *Fine-tuning* [Bai et al., 2022, Ouyang et al., 2022] demands per-culture labelled data and modifies weights irreversibly. *Prompting-based steering*

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

relies on ad-hoc instructions with no empirical grounding [Jin et al., 2025]. *Activation steering* [Arditi et al., 2025, Zou et al., 2023] requires white-box access and per-model engineering. *Reward-guided decoding* [Khanov et al., 2024, Mudgal et al., 2024] needs trained reward models unavailable for cross-cultural settings. *Culture-aware fine-tuning* [Li et al., 2024, Yao et al., 2024] trains adapters per culture but scales poorly across 100+ countries.

We propose **SWA-MPPI** (Socially-Weighted Alignment with Model Predictive Path Integral control), grounded in three observations. First, moral preferences within any society are not monolithic: age cohorts and social strata exhibit systematically different priorities [Awad et al., 2018, Kleiman-Weiner et al., 2017]. Second, the World Values Survey [Haerpfer et al., 2022] provides empirically calibrated data on these within-country differences across 80+ nations. Third, when culturally grounded persona agents *disagree* on a decision, that disagreement is an informative signal exploitable through stochastic optimal control [Williams et al., 2018].

SWA-MPPI operates as follows. For a target country, it constructs $N = 4$ system-prompt personas from WVS Wave 7 statistics (three age cohorts plus a utilitarian agent), evaluates all $N + 1$ agents in a single batched forward pass, and extracts decision logits. If inter-agent variance exceeds a calibrated threshold τ, it samples $K = 128$ perturbations and scores each via a Prospect Theory value function [Kahneman and Tversky, 1979]. The importance-weighted MPPI update yields a minimal-KL correction; a positional debiasing step cancels recency effects.

Contributions.

- We introduce SWA-MPPI, the first application of MPPI in logit space for cross-cultural value alignment: a training-free framework combining WVS-grounded personas, adaptive conflict detection, and Prospect-Theory-weighted optimisation (§3).
- We demonstrate mean JSD reductions of **37.6%** (Qwen2.5-72B) and **34.7%** (Llama-3.1-70B) across 15 countries, with gains reaching $+75\%$ for Anglosphere pairs, outperforming all baselines by $\geq 10\%$ (§5).
- We conduct systematic ablations revealing a clear component hierarchy: the utilitarian persona is most critical (-39.1% relative Pearson without it), followed by MPPI ($+15.9\%$ JSD), debiasing ($+12.0\%$), and Prospect Theory ($+4.5\%$) (§6).

2 Related Work

Cross-cultural moral benchmarks. The Moral Machine experiment [Awad et al., 2018] provides 40M+ cross-cultural moral judgments. MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs—we adopt it as our benchmark and propose an alignment *intervention* rather than diagnostics alone. Other benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric. PRISM [Kirk et al., 2024] and GlobalOpinionQA [Durmus et al., 2023] address diversity but cover few languages. Recent frameworks (EvalMORAAL [Aakanksha et al., 2024], CARB [Myung et al., 2024]) emphasise broader-topic evaluations; SWA-MPPI targets binary moral decisions specifically.

Cross-cultural LLM evaluation. LLM outputs correlate most strongly with liberal, educated American respondents [Santurkar et al., 2023]. RLHF introduces unintended cultural impacts [Ryan et al., 2024], and assigning cultural personas can surface toxic outputs [Deshpande et al., 2023]. This recurring WEIRD bias motivates SWA-MPPI's calibration objective.

Inference-time alignment. Activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access. ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] need trained reward models—unavailable for cross-cultural settings (Appendix G). CultureManager [Li et al., 2024] and CAReDiO [Yao et al., 2024] train culture-specific adapters but require per-culture labelled data and weight modification, scaling poorly across 100+ countries. SWA-MPPI is the first *training-free* method addressing multi-cultural calibration. Sorensen et al. [2024] articulate the goal of pluralistic alignment, which SWA-MPPI operationalises concretely.

MPPI and Prospect Theory. MPPI [Williams et al., 2018] is a sampling-based stochastic optimal control algorithm originally developed for robotics; Huang et al. [2024] adapted it for token-level

LLM steering. We are the first to apply MPPI in *logit space* for moral alignment, combining it with Prospect Theory [Kahneman and Tversky, 1979] to capture asymmetric gain-loss sensitivity in human moral judgment.

3 Method: SWA-MPPI

3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `LEFT` (ℓ) and `RIGHT` (r). For target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [Awad et al., 2018]. A frozen LLM f_θ produces logits $\mathbf{z}(\mathbf{x}) = [z_\ell,\ z_r]$; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_r - z_\ell$. Each dimension's AMCE is estimated as:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention (see Appendix A for consistency justification).

Goal: Without modifying θ, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$ under Jensen-Shannon Divergence.

3.2 Cultural Persona Construction

For each country c, SWA-MPPI constructs $N = 4$ persona agents from WVS Wave 7 data [Haerpfer et al., 2022] across 8 value dimensions (gender egalitarianism, religious importance, interpersonal trust, moral permissiveness, work centrality, family importance, personal autonomy, meritocratic orientation). Three personas correspond to age cohorts (*young* 18–35, *middle-aged* 36–55, *older* >55), each parameterised by country-specific WVS statistics. A fourth, country-invariant *utilitarian agent* maximises lives saved. Each persona is a natural-language system prompt:

> *You are a [age-cohort] adult from [country]. The cultural values typical of your social group are: [WVS statistics]. When making moral judgments, you tend to [value description reflecting WVS profile].*

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix B.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures T_{cat} (Table 1) scale logits before computing decision gaps:

$$\delta_i = \frac{z_{i,r} - z_{i,\ell}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N} \sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

The reward r_i measures how strongly persona i shifts the model *away from the culturally-neutral baseline*—a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit.

3.4 Adaptive Conflict Detection

When personas agree, the consensus $\bar{\delta}$ suffices. When they disagree substantially, a controlled correction is warranted. SWA-MPPI uses a per-country threshold $\tau^{(c)}$ calibrated on $n_{\text{cal}} = 50$ unlabelled scenarios to target a 35% MPPI trigger rate:

$$\tau^{(c)} = \text{Percentile}_{65}\Big(\big\{\text{Var}(\delta_1^{(s)}, \ldots, \delta_N^{(s)})\big\}_{s=1}^{n_{\text{cal}}}\Big). \tag{5}$$

Table 1: SWA-MPPI hyperparameters. PT parameters from Kahneman and Tversky [1979]; others validated on a held-out synthetic set (Appendix D).

Parameter	Symbol	Value	Source
Persona agents	N	4	Design
MPPI samples	K	128	Design
Perturbation σ	σ	0.3	Validation
Cooperation weight	λ_{coop}	0.7	Validation
KL penalty	α_{KL}	0.05	Validation
PT curvature	$\alpha{=}\beta$	0.88	Kahneman and Tversky [1979]
PT loss aversion	κ	2.25	Kahneman and Tversky [1979]
Decision temp.	T_{dec}	0.5	Validation
Trigger rate	ρ_{target}	0.35	Design
T_{cat} (Species / Gender / Others)		4.0 / 3.5 / 1.5	Category

MPPI fires iff $\mathrm{Var}(\delta_1, \ldots, \delta_N) \geq \tau^{(c)}$. Calibration uses only variance statistics and never observes human AMCEs (Appendix E).

3.5 Prospect-Theory-Weighted MPPI Optimisation

When triggered, SWA-MPPI draws $K = 128$ perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ ($\sigma = 0.3$) and evaluates a collective welfare score for each $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$.

Per-agent cooperative utility.

$$U_i(\epsilon_k) = (1 - \lambda_{\text{coop}})\, v(r_i\, \tilde{\delta}_k) \;+\; \lambda_{\text{coop}}\, v\Big(\bar{r}_{-i}\, \tilde{\delta}_k\Big), \tag{6}$$

where $\lambda_{\text{coop}} = 0.7$ and $v(\cdot)$ is the Prospect Theory value function with $\alpha = \beta = 0.88$, $\kappa = 2.25$ [Kahneman and Tversky, 1979]:

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0, \\ -\kappa\, |x|^{\beta} & x < 0. \end{cases} \tag{7}$$

Collective utility with KL regularisation.

$$U_{\text{total}}(\epsilon_k) = \frac{1}{N} \sum_{i=1}^{N} U_i(\epsilon_k) \;-\; \alpha_{\text{KL}}\, \frac{\epsilon_k^2}{2\sigma^2}, \tag{8}$$

where $\alpha_{\text{KL}} = 0.05$. The KL term derives from the free-energy objective of MPPI, preventing overcorrection in a theoretically grounded way (Appendix C).

MPPI update.

$$w_k = \frac{\exp\big(U_{\text{total}}(\epsilon_k)/\beta\big)}{\sum_{k'} \exp\big(U_{\text{total}}(\epsilon_{k'})/\beta\big)}, \qquad \delta^* = \sum_{k=1}^{K} w_k\, \epsilon_k. \tag{9}$$

3.6 Positional Debiasing and Final Prediction

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice with swapped group orderings; decision gaps are averaged before consensus computation. The final prediction is $p_{\text{spare}} = \sigma(\delta_{\text{opt}}/T_{\text{dec}})$ where $\delta_{\text{opt}} = \bar{\delta} + \delta^*$ and $T_{\text{dec}} = 0.5$.

Algorithm 1 summarises the full procedure; Table 1 lists all hyperparameters (validation details in Appendix D).

4 Experimental Setup

4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 2). For each of 15 countries, we apply deduplication, quality filtering,

Algorithm 1 SWA-MPPI Inference for Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^N$; threshold $\tau^{(c)}$; hyperparameters $K, \sigma, \lambda_{\text{coop}}, \alpha_{\text{KL}}, T_{\text{dec}}$
Ensure: $p_{\text{spare}} \in [0, 1]$

```
for order ∈ {orig, swap} do
    Batched forward: evaluate f_θ with base + N personas
    Extract logit pairs; compute δ_i^(ord)                    ▷ Eq. 2
end for
Average: δ_i ← (δ_i^(orig) − δ_i^(swap))/2                    ▷ Debiasing
Compute consensus δ̄, rewards r_i, variance ν                 ▷ Eqs. 3–4
if ν ≥ τ^(c) then                                             ▷ Conflict detected
    for k = 1 to K do
        Sample ε_k ~ N(0, σ²); compute U_total(ε_k)           ▷ Eqs. 6–8
    end for
    δ* = Σ_k w_k ε_k                                          ▷ Eq. 9
else  δ* ← 0
end if
return p_spare = σ((δ̄ + δ*)/T_dec)
```

Figure placeholder: SWA-MPPI Pipeline Overview

Left: WVS Persona Construction → Middle: Batched Forward Pass Loop → Right: MPPI Optimisation

Figure 1: **SWA-MPPI pipeline. Left:** WVS-grounded persona construction (3 age cohorts + utilitarian). **Middle:** Batched forward pass with positional debiasing. **Right:** MPPI optimisation when inter-persona variance exceeds $\tau^{(c)}$.

category capping (max 80), and up-sampling (min 36 per dimension), yielding 310–500 scenarios per country (Appendix E). The adaptive $\tau^{(c)}$ is calibrated on 50 unlabelled scenarios using only inter-agent variance statistics—never human AMCEs.

4.2 Models

Six LLMs: **Qwen2.5-72B-Instruct** [Qwen Team, 2025] (Int8), **Llama-3.1-70B-Instruct** [Grattafiori et al., 2024] (FP8), **Qwen2.5-7B/32B-Instruct** and **Llama-3.1-8B-Instruct** (4-bit), **Mistral-Large-2407** [Mistral AI Team, 2024] (4-bit). All on a single H100 80GB via Unsloth [Unsloth AI, 2024], greedy decoding, seed 42. Scenarios use native-language bodies with English decision tokens (Appendix F).

Table 2: The six moral dimensions in MultiTP.

Dimension	Preferred	Contra	T_{cat}
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

Table 3: Cross-model comparison. Bootstrap 95% CIs; differences $>\pm 0.004$ significant at $p<0.05$.

Model	Size	Mean JSD ↓		Improv. (%)	Pearson r ↑	
		Van.	SWA		Van.	SWA
Qwen2.5-72B	72B	$.0859 \pm .005$	$.0536 \pm .003$	**+37.6**	0.548	**0.696**
Llama-3.1-70B	70B	$.0937 \pm .006$	$.0612 \pm .004$	+34.7	$-.026$	0.384
Llama-3.1-8B	8B	$.0533 \pm .004$	$.0436 \pm .003$	+22.6	$-.247$	0.271
Qwen2.5-7B	7B	$.0522 \pm .004$	$.0426 \pm .003$	+18.4	0.252	0.402
Mistral-Large	126B	$.0457 \pm .003$	$.0430 \pm .003$	+5.8	0.658	0.550
Qwen2.5-32B	32B	$.0632 \pm .004$	$.0633 \pm .004$	-0.2	0.625	0.550

4.3 Baselines

On Llama-3.1-70B: (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [Kirk et al., 2024], (5) **Activation Steering** [Arditi et al., 2025] using culturally contrastive steering vectors (details in Appendix G). Reward-guided methods (ARGS, controlled decoding) are excluded as they require reward models unavailable for cross-cultural settings (Appendix G).

4.4 Metrics

JSD (primary, ↓): Jensen-Shannon Distance between model and human AMCE vectors. **Pearson** r (↑): correlation of 6D AMCE vectors. **MAE** (↓): mean absolute AMCE error. All aggregated as unweighted means across 15 countries with bootstrap 95% CIs (1000 resamples).

5 Results

5.1 Cross-Model Comparison

Table 3 summarises performance across six models. The two 70B-class models benefit most: Qwen2.5-72B achieves **37.6%** mean JSD improvement (Pearson r: $0.548 \rightarrow 0.696$) and Llama-3.1-70B achieves **34.7%** (r: $-0.026 \rightarrow 0.384$). Smaller models show consistent +18–23% gains. Mistral-Large (+5.8%) has high baseline alignment leaving less room for correction. Qwen2.5-32B (-0.2%) exhibits concentrated logit distributions causing uninformative MPPI weights—an adaptive σ policy could address this (Appendix K).

5.2 Baseline Comparison on Llama-3.1-70B

Table 4: SWA-MPPI achieves the lowest mean JSD (0.0612) and highest Pearson r (0.384), outperforming the best alternative (country-tailored prompting) by 10.0%. Activation steering performs *below* vanilla, suggesting contrastive steering vectors interfere with moral reasoning on this benchmark.

5.3 Per-Country Results: Qwen2.5-72B

Table 5: SWA-MPPI improves JSD in 13/15 countries. **The Anglosphere paradox:** the largest gains occur for Australia (+75.4%), Great Britain (+72.2%), and USA (+71.1%)—precisely the English-

Table 4: Baseline comparison on Llama-3.1-70B across 15 countries.

Method	Mean JSD ↓	Pearson r ↑	ΔJSD vs SWA
SWA-MPPI (ours)	$\mathbf{.0612 \pm .004}$	**0.384**	–
Country-Tailored (B1)	$.0680 \pm .004$	0.233	+11.1%
PRISM-Style (B3)	$.0699 \pm .005$	0.148	+12.5%
WVS Profile (B2)	$.0737 \pm .005$	0.155	+16.9%
Activation Steering	$.0877 \pm .006$	−0.208	+30.2%
Vanilla	$.0937 \pm .006$	−0.026	+34.7%

Table 5: Per-country results for **Qwen2.5-72B-Instruct**.

Country	Lang.	JSD ↓		Pearson r ↑		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.1066	**.0308**	0.400	0.822	−.076	+71.1
Germany	de	.0603	.0603	0.845	0.747	+.000	+0.0
China	zh	.0513	.0371	0.851	0.762	−.014	+27.8
Japan	ja	.0986	.0553	0.634	0.786	−.043	+43.9
Brazil	pt	.0754	.0703	0.349	0.529	−.005	+6.7
Saudi Arabia	ar	.0778	.0796	0.443	0.527	+.002	−2.3
Vietnam	vi	.1302	.0783	0.569	0.456	−.052	+39.9
France	fr	.0789	.0479	0.617	0.838	−.031	+39.4
India	hi	.0600	.0561	0.670	0.739	−.004	+6.5
South Korea	ko	.0720	.0513	0.355	0.645	−.021	+28.8
Great Britain	en	.1057	**.0294**	0.419	0.800	−.076	+72.2
Russia	ru	.0800	.0629	0.623	0.784	−.017	+21.4
Mexico	es	.0760	.0534	0.605	0.545	−.023	+29.8
Nigeria	en	.1103	.0649	0.407	0.615	−.046	+41.2
Australia	en	.1047	**.0258**	0.438	0.841	−.079	+75.4
Mean		**.0859**	**.0536**	0.548	0.696	−.032	**+37.6**

speaking populations one might expect to be best served. Their high vanilla JSD (≈0.10–0.11) reflects RLHF-induced biases (extreme animal protection, strong gender-neutral stances) that diverge from average human preferences.

Germany (ceiling case): vanilla JSD = 0.0603 with $r = 0.845$—already exceptionally aligned. SWA-MPPI holds steady (JSD unchanged), demonstrating its *self-limiting* property: low inter-persona disagreement means the trigger rarely fires. The Pearson r decrease (0.845→0.747) reflects sensitivity to small rank perturbations in 6D, not distributional degradation (Appendix J).

Saudi Arabia (−2.3%) and Brazil (+6.7%) have limited WVS Wave 7 coverage, constraining persona quality.

6 Ablation Study

We ablate six components on Qwen2.5-72B, USA (342 scenarios; Table 6). Note: the ablation uses the full scenario pool, yielding JSD = 0.0425 vs. 0.0308 in Table 5 due to different dataset splits (Appendix H).

The results establish a clear hierarchy: (i) the **utilitarian persona** is most critical (+43.5% JSD without it; Pearson r collapses by 0.283), anchoring the ensemble along the numerosity dimension; (ii) **MPPI** adds meaningful refinement (+15.9% JSD); (iii) **positional debiasing** corrects substantial recency bias (+12.0% JSD; Species preference inflated by 7.7pp without it); (iv) **Prospect Theory** contributes through nonlinear loss-averse weighting (+4.5% JSD vs. linear). The adaptive trigger fires at 100% for USA—a genuine empirical property of high-disagreement scenarios, not a calibration failure. The MPPI *flip rate* is only 6.0%, indicating the framework *modulates confidence* rather than reversing decisions (Appendix I).

Table 6: Ablation on Qwen2.5-72B, USA (342 scenarios). Full system: JSD = 0.0425, r = 0.724.

#	Configuration	JSD ↓	ΔJSD	r ↑	Δr	MAE
	Full SWA-MPPI	**0.0425**	–	**0.724**	–	**9.77**
1	No-MPPI (consensus only)	0.0493	+.007	0.650	−.074	11.02
2	Always-on MPPI	0.0422	−.000	0.729	+.005	9.74
3a	Utility: quadratic	0.0429	+.000	0.725	+.001	9.68
3b	Utility: linear	0.0444	+.002	0.706	−.018	10.12
4	Without utilitarian (N=3)	0.0610	+.019	0.441	−.283	11.09
5	No debiasing	0.0476	+.005	0.743	+.019	9.67

7 Discussion

Key findings. SWA-MPPI achieves three desirable properties simultaneously: (i) substantial alignment gains on well-calibrated 70B models without weight modification, (ii) graceful self-limiting when intervention is unnecessary (Germany), and (iii) a 4.6× latency overhead (730ms vs. 160ms per scenario on H100) negligible for batch inference. The MPPI loop operates on cached scalars and contributes negligibly to wall-clock time.

The Anglosphere paradox. The largest gains occur for English-speaking countries—precisely those best-represented in training data. This reveals that RLHF introduces specific moral biases (extreme animal protection, strong gender-neutral stances) that *actively worsen* alignment with even Western human preferences, and SWA-MPPI's WVS-derived personas successfully counterbalance these.

Limitations. We evaluate 15 of 100+ MultiTP countries; extending to low-resource languages remains untested. The per-agent reward r_i centres corrections relative to the base model, encoding an assumption that persona shifts approximate human directions—validated at aggregate level but unverified per-dimension. The WVS grounding encodes surveyed averages at a specific historical moment (Wave 7, ∼2017–2022); temporal staleness and limited WVS coverage for some countries constrain applicability. Extended limitations are in Appendix K.

Broader impact. SWA-MPPI could enable more culturally equitable AI but also risks aligning to harmful population-level preferences. WVS grounding provides empirical accountability; deployment should include governance oversight and periodic persona updates.

8 Conclusion

We presented SWA-MPPI, a training-free inference-time framework for cross-cultural moral alignment. By grounding persona construction in WVS data, exploiting inter-persona disagreement as a trigger signal, and applying Prospect-Theory-weighted MPPI in logit space, SWA-MPPI reduces JSD to human preferences by 34–38% on 70B models, outperforms all baselines, and gracefully refrains from correction when alignment is already high. The Anglosphere paradox illustrates how RLHF biases worsen alignment even for well-represented populations. Future directions include adaptive σ for broader model compatibility, extension to multi-option settings, and integration with newer WVS waves.

Acknowledgments and Disclosure of Funding

We thank the creators of MultiTP [Jin et al., 2025] and the Moral Machine experiment [Awad et al., 2018] for public data. Experiments on NVIDIA H100 hardware.

References

Aakanksha et al. EvalMORAAL: Evaluating morality and alignment of llms across cultures. *arXiv preprint arXiv:2410.01999*, 2024.

A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, volume 37, pages 136037–136083, 2025.

E. Awad, S. Dsouza, R. Kim, J. Schulz, J. Henrich, A. Shariff, J.-F. Bonnefon, and I. Rahwan. The moral machine experiment. *Nature*, 563(7729):59–64, 2018.

Y. Bai, S. Kadavath, S. Kundu, A. Askell, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

A. Deshpande, V. Murahari, T. Rajpurohit, A. Kalyan, and K. Narasimhan. Toxicity in ChatGPT: Analyzing persona-assigned language models. *arXiv preprint arXiv:2304.05335*, 2023.

E. Durmus, K. Nguyen, T. I. Liao, N. Schiefer, A. Askell, A. Bakhtin, et al. Towards measuring the representation of subjective global opinions in language models. *arXiv preprint arXiv:2306.16388*, 2023.

A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

C. Haerpfer, R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, and B. Puranen. World Values Survey: Round seven – country-pooled datafile. Madrid, Spain & Vienna, Austria: JD Systems Institute & WVSA Secretariat, 2022. `https://doi.org/10.14281/18241.20`.

D. Hendrycks, C. Burns, S. Basart, A. Critch, J. Li, D. Song, and J. Steinhardt. Aligning AI with shared human values. In *International Conference on Learning Representations*, 2021.

J. Henrich, S. J. Heine, and A. Norenzayan. The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2-3):61–83, 2010.

Z. Huang, W. Chen, C. Lyu, et al. MPPI-guided large language model generation. *arXiv preprint arXiv:2403.09269*, 2024.

Z. Jin, S. Levine, F. Gonzalez Adauto, O. Kamal, M. Sap, M. Sachan, R. Mihalcea, J. Tenenbaum, and B. Schölkopf. When to make exceptions: Exploring language models as accounts of human moral judgment. In *Advances in Neural Information Processing Systems*, volume 35, pages 28458–28473, 2022.

Z. Jin, M. Kleiman-Weiner, G. Piatti, S. Levine, J. Liu, F. Gonzalez, F. Ortu, A. Strausz, M. Sachan, R. Mihalcea, Y. Choi, and B. Schölkopf. Language model alignment in multilingual trolley problems. *arXiv preprint arXiv:2407.02273*, 2025. Published as a conference paper at ICLR 2025.

D. Kahneman and A. Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.

M. Khanov, J. Burapacheep, and Y. Li. ARGS: Alignment as reward-guided search. *arXiv preprint arXiv:2402.01694*, 2024.

H. R. Kirk, A. Whitefield, P. Röttger, A. Bean, et al. The PRISM alignment project: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. *arXiv preprint arXiv:2404.16019*, 2024.

M. Kleiman-Weiner, R. Saxe, and J. B. Tenenbaum. Learning a commonsense moral theory. *Cognition*, 167:107–123, 2017.

Y. Li et al. Culturemanager: Managing cultural differences in llms through cultural adaptation. *arXiv preprint arXiv:2409.02945*, 2024.

N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

Mistral AI Team. Mistral NeMo. *arXiv preprint arXiv:2407.xxxxx*, 2024. Mistral Large 2 (Instruct-2407) model card.

S. Mudgal, J. Lee, H. Ganapathy, Y. Li, T. Wang, Y. Huang, Z. Chen, H.-T. Cheng, M. Collins, T. Strohman, et al. Controlled decoding from language models. *arXiv preprint arXiv:2310.17022*, 2024.

J. Myung et al. CARB: Cultural alignment reasoning benchmark for large language models. *arXiv preprint arXiv:2406.14766*, 2024.

A. Nie, Y. Zhang, A. Amdekar, C. Piech, T. B. Hashimoto, and T. Gerstenberg. MoCa: Measuring human-language model alignment on causal and moral judgment tasks. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

L. Ouyang, J. Wu, X. Jiang, D. Almeida, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.

M. J. Ryan, W. Held, and D. Yang. Unintended impacts of LLM alignment on global representation. *arXiv preprint arXiv:2402.15018*, 2024.

S. Santurkar, E. Durmus, F. Ladhak, C. Lee, P. Liang, and T. Hashimoto. Whose opinions do language models reflect? *arXiv preprint arXiv:2303.17548*, 2023.

T. Sorensen, J. Moore, J. Fisher, M. Gordon, N. Mireshghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024.

Unsloth AI. Unsloth: Fast LLM fine-tuning and inference, 2024. URL `https://github.com/unslothai/unsloth`.

G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou. Information-theoretic model predictive control: Theory and applications to autonomous driving. In *IEEE Transactions on Robotics*, volume 34, pages 1603–1622. IEEE, 2018.

J. Yao et al. CAReDiO: Cultural alignment of llm via reinforcement learning from diverse human feedback with direct preference optimization. *arXiv preprint arXiv:2410.18203*, 2024.

A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A AMCE Estimation Consistency

We estimate each dimension's AMCE as the marginal preference rate (MPR, Eq. 1), consistent with MultiTP conventions [Jin et al., 2025]. The human ground-truth AMCEs in `country_specific_ACME.csv` are derived from the same marginal averaging over 40M+ Moral Machine responses, ensuring comparable computation. A full conjoint regression estimator would correct for co-occurrence patterns; however, Awad et al. [2018] show that marginal averages are unbiased under the balanced randomisation design, making Eq. 1 valid.

B Persona Construction Details

B.1 Example Personas

The young-adult persona for the **United States**:

> *You are a young adult (18–35) from the United States of America. Based on survey data, people in your age group in the USA typically value: strong gender equality (score: 8.2/10), low religious centrality in moral decisions (score: 3.1/10), high personal autonomy (score: 7.8/10), high meritocratic orientation (score: 7.5/10), moderate family importance (score: 6.4/10), and moderate interpersonal trust (score: 5.3/10). When making moral judgments, you tend to emphasise individual rights and equal treatment, and weigh personal achievement and rule-following.*

The older-adult persona for **Vietnam**:

> *You are an older adult (55+) from Vietnam. Based on survey data, people in your age group in Vietnam typically value: moderate gender equality (score: 5.1/10), high religious and moral traditionalism (score: 7.2/10), strong family importance (score: 9.4/10), moderate personal autonomy (score: 4.8/10), and low interpersonal trust toward strangers (score: 3.2/10). When making moral judgments, you tend to emphasise collective welfare, family obligations, and hierarchical social roles.*

The utilitarian persona (country-invariant): *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

B.2 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

Direct thematic overlap: WVS *gender egalitarianism* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religious importance* correlates with Species (human exceptionalism); *meritocratic orientation* relates to Social Value and Fitness.

Indirect modulation: Dimensions like *personal autonomy*, *interpersonal trust*, and *family importance* modulate the moral *frame*—e.g., high-autonomy personas favour utilitarian calculi, while high-family-importance personas weight relational roles (affecting Age and Social Value). Crucially, SWA-MPPI does not require exact causal mapping: the ensemble generates diversity in logit-space gaps, and MPPI selects the correction balancing collective utility. A rigorous per-dimension causal analysis (e.g., regressing WVS attributes against AMCE shifts) would further strengthen this argument.

C MPPI Theoretical Justification

C.1 Degenerate but Principled Stochastic Optimal Control

The scalar decision gap δ constitutes a one-dimensional *state*; "dynamics" are identity (no state transition); perturbations ϵ_k are "control inputs" at a single horizon step. This is a degenerate case of MPPI [Williams et al., 2018], but the information-theoretic derivation remains valid: $w_k \propto \exp(U_{\text{total}}/\beta)$ minimises KL divergence between the sample distribution and the Boltzmann distribution of utility, making the KL term in Eq. 8 a direct consequence of the free-energy objective—not a heuristic penalty.

C.2 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. MPPI performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. The ablation (Table 6) provides indirect evidence: replacing PT with linear utility (closer to uniform weighting, hence a simple mean shift) degrades JSD by 4.5%.

D Hyperparameter Validation

Non-PT hyperparameters (σ, λ_{coop}, α_{KL}, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures.

E Dataset Preprocessing and Calibration

E.1 Preprocessing Sensitivity

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. A formal sensitivity analysis (varying capping ratios, up-sampling rates) is recommended for future work.

E.2 Calibration Limitation

The 50-scenario $\tau^{(c)}$ calibration set is drawn from the evaluation distribution. We mitigate overfitting risk: (i) calibration observes only a single scalar (variance), unlikely to overfit; (ii) it targets a fixed trigger rate, not alignment metrics; (iii) the 100% trigger rate for USA suggests τ is not sensitively tuned. A fully separate calibration corpus would provide stronger guarantees.

F Cross-Lingual Token Elicitation

Scenarios use native-language bodies with English decision suffix: `Which group should be spared? Answer with LEFT or RIGHT only.` Token IDs verified per model at initialisation (`add_special_tokens=False`): Qwen2.5: LEFT=22397, RIGHT=11572; Llama-3.1: LEFT=31266, RIGHT=9268. Logits extracted from last-position output tensor; no generation sampling performed.

G Baseline Implementation Details

G.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs. Vectors are extracted from layer 32 (transformer midpoint) and applied with scaling $\alpha = 1.5$, selected via the synthetic validation set.

G.2 Omitted Baselines

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] require trained reward models. In the cross-cultural setting, this demands either a separate reward model per country or culturally conditioned models—neither exists for MultiTP. SWA-MPPI's constraint is operating without labelled preference data, making these methodologically incompatible.

H Detailed Ablation Analysis

JSD difference with Table 5: The ablation uses the full 342-scenario pool; per-country results use the evaluation split after calibration reservation, yielding different scenario compositions.

Always-on vs. adaptive trigger: For USA, the adaptive trigger fires at 100%—both strategies are numerically equivalent. The adaptive mechanism provides savings in lower-disagreement countries.

Utilitarian persona: Removing it collapses Utilitarianism MPR from 61.2% to 48.0% (human: 76.6%), confirming it anchors the numerosity dimension.

Debiasing: Mean positional bias is 0.193 logit units; Species preference inflates from 85.8% to 93.5% without it (human: 79.2%).

I Trigger Mechanism and Latency

The adaptive τ targets 35% activation but achieves 100% for USA (all scenarios contested). The *flip rate* (MPPI reversals) is 6.0%, indicating confidence modulation rather than decision reversal.

Mean latency: 730ms (SWA-MPPI) vs. 160ms (vanilla) on Qwen2.5-72B (H100). Overhead from batched multi-agent passes and debiasing; MPPI operates on cached scalars and contributes negligibly.

J Germany Pearson r Analysis

Germany's JSD remains unchanged (0.0603) while Pearson r drops from 0.845 to 0.747. In 6D, Pearson r is sensitive to rank perturbations: a single dimension swap reduces r by $\sim$0.1 while JSD (depending on magnitudes, not ordering) stays constant. Rare MPPI corrections introduce minor rank perturbations without affecting distributional distance.

K Extended Limitations

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation is needed.

Country coverage. 15 countries across major cultural clusters, but no low-resource languages (Amharic, Khmer, Yoruba). Cross-lingual generalisation to such settings remains untested.

Binary format. The logit-difference formulation requires generalisation to softmax for multi-option settings.

Language–culture conflation. One language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting MPPI stability.

WVS temporal staleness. Wave 7 ($\sim$2017–2022) data may not reflect current preferences in rapidly evolving societies.

Qwen2.5-32B failure mode. Concentrated logit distributions cause MPPI perturbations to be off-manifold, yielding near-uniform importance weights. Adaptive σ based on per-model logit entropy could address this.